

Devansh Pancholi

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PROFILE SUMMARY

Computational biologist with hands-on experience in cancer genomics, plasma cfDNA liquid-biopsy analysis, epigenomics, low-pass WGS fragmentomics, single-cell and bulk transcriptomics, somatic-variant analysis, tumor-immune state analysis, cancer biomarker evaluation, regulatory genomics, disease-progression analysis, and clinical cohort analysis. Proficient in Python, R, C++, Linux/HPC, Snakemake, Nextflow, Docker, AWS, machine learning, and scalable NGS workflow development, with additional experience in RNA biology and AI/deep-learning-enabled RNA structure prediction.

EDUCATION

Master of Science in Bioinformatics | Saint Louis University, USA

May 2026

Bachelor of Engineering in Biomedical Engineering | L.D. College of Engineering, India

June 2023

TECHNICAL SKILLS

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Programming & Computing: Python, R, C++, Bash/Shell, SQL, Linux/UNIX, HPC, SLURM, AWS, Web Development

Genomics & Transcriptomics: NGS Analysis, WGS, cfDNA, fragmentomics, epigenomics, bulk RNA-seq, scRNA-seq, differential expression, pathway analysis, tumor-immune/cell-state analysis, somatic-variant analysis, biomarker discovery

Tools & Workflows: BWA, SAMtools, GATK, STAR, Bowtie2, Scanpy, IGV, UCSC Genome Browser, Clustal Omega, BLAST, featureCounts, DESeq2/edgeR, Docker, Nextflow, Snakemake

Machine Learning & AI: Scikit-learn, XGBoost, SHAP, GNNs, diffusion models, transformer encoders, RiNALMo, ESM-2

Structural & Data Analysis: RMSD, TM-score, Kabsch alignment, RNA motif modelling, PyMOL, Chimera, Matplotlib, ggplot, Jupyter

RESEARCH EXPERIENCE

Bioinformatics Analyst | Saint Louis University | **Advisor: Dr. Ted**

April 2026 – Present

HNSCC Tumor-Immune State & Multi-Modal RNA Biomarker | Immuno-Oncology

- Analysing human HNSCC single-cell RNA-seq datasets to characterize tumor and immune-cell microenvironments through QC, dimensionality reduction, cell-type annotation, differential expression, and pathway/module scoring, to understand tumor, lymphoid, and myeloid populations, with emphasis on tumor-associated macrophage and immune-response states.
- Defining and evaluating macrophage-state signatures centered on SPP1-enriched immunosuppressive programs versus CXCL9/CXCL10 responsive immune-active programs, while assessing antigen presentation, inflammatory signaling, angiogenesis, matrix remodeling, and T-cell-dysfunction-associated pathways across independent HNSCC datasets.
- Integrating M-I treatment-response single-cell data with HNSCC macrophage signatures to test reprogramming from SPP1-enriched immunosuppressive states toward CXCL9/CXCL10 inflammatory/interferon-responsive programs, using treated-versus-control analysis, pathway scoring, human-mouse ortholog mapping for conserved immune-response interpretation.
- Extending single-cell-derived immune-state signatures into multi-modal and cohort-level analysis using TCGA-HNSC bulk transcriptomics, macrophage-state scoring/deconvolution, and clinical variables to evaluate associations with tumor biology.
- Translating reproducible immune-state signatures toward non-invasive biomarker and therapeutic discovery by integrating GTEx salivary/mucosal expression, exRNA detectability, compact RNA-panel prioritization, and LINCS L1000/Connectivity Map reversal analysis to identify perturbations predicted to oppose SPP1-rich immunosuppressive programs.

Research Assistant (WashU) | **Advisor: Dr. Maher**

Jan 2026- May 2026

Epigenomics Biomarker Discovery for Colorectal Cancer | Washington University in Saint Louis

- Conducted sequencing-based liquid-biopsy biomarker research integrating plasma cfDNA 5hmC profiling and low-pass whole-genome sequencing (LP-WGS) fragmentomics across discovery and independent validation cohorts to characterise colorectal cancer biology and evaluate progression-related molecular signals.

- Implemented, optimized, and troubleshooted reproducible Python/R and Linux/HPC workflows for alignment, quality control, gene-body 5hmC quantification, fragment-feature generation, regulatory analysis, and cohort-scale validation using BWA-MEM, SAMtools, Docker/LSF, MACS2, featureCounts, DESeq2/edgeR, and Griffin.
- Characterised cancer-associated fragmentation and chromatin organisation through fragment-length distributions, short/long fragment ratios, 4-mer and 5-mer end motifs, nucleosome occupancy, and transcription-factor-binding-site accessibility inference.
- Analysed baseline/pre-surgery cfDNA profiles in relation to later two-year clinical outcomes, identifying progression-associated enrichment of several genes and supporting the potential of cfDNA 5hmC for recurrence-risk and survival-related biomarker research.
- Evaluated the effects of EDTA versus Streck blood-collection conditions and other pre-analytical variability on fragment-length and end-motif profiles, strengthening quality-control and biomarker-interpretation strategies.

Research Assistant (SLU) | Advisor: Dr. Hou

Mar 2025 – May 2026

AI and Deep-Learning Driven RNA Structure Prediction | Saint Louis University

- Developed Python/C++ workflows for RNA motif extraction, sequence/structure alignment, RMSD clustering, template search, and template-guided 3D structure assembly; built RNAMotifDB integrating sequence, secondary-structure, and tertiary-template information.
- Extended GNN- and diffusion-based prediction frameworks with RiNALMo embeddings, motif-aware representations, and residue-level alignment, while implementing structural repair and coordinate mapping for missing residues, variable-length inputs, and sequence–structure inconsistencies.
- Designed parallel HPC/Nextflow pipelines for preprocessing, template extraction, model inference, PDB generation, and evaluation; benchmarked OpenFold3/Boltz-style template and synthetic-MSA strategies across 30–238-nt RNAs, improving structural consistency and reducing RMSD by up to ~35%.

Publications & Technical Projects

SELECTED CANCER GENOMICS PROJECTS

- **Breast Cancer Single-Cell Atlas | Scanpy, scRNA-seq, Differential Expression, Pathway Analysis**
 - Integrated 72 breast cancer scRNA-seq samples across normal, ER+, HER2+, TNBC, and metastatic cohorts to build a unified atlas and characterize tumor/microenvironment cell populations, subtype-specific states, differential transcriptional programs, and pathway activity.
[GitHub: Breast Cancer Single-Cell Atlas](#)
- **Breast Cancer Somatic Variant Analysis | WGS, GATK Mutect2, SnpEff, Cancer Genomics**
 - Built a reproducible tumor-normal WGS pipeline using BWA-MEM, SAMtools, GATK Mutect2, and SnpEff for somatic variant detection, filtering, annotation, driver prioritization, and pathway-level interpretation.
[GitHub: Breast Cancer Somatic Mutation Analysis](#)
- **Smart Stethoscope - IoT-enabled Medical Device | IEEE ASIANCON 2023 ([IEEE Xplore](#))**